LearnFlow

A modern web-based Learning Management System bridging teachers and students for seamless online assessments

Project Overview

LearnFlow is a comprehensive web-based platform designed to transform how educators create, deliver, and track assessments. Built with security and usability at its core, the system provides teachers with powerful tools to manage classes and quizzes while offering students an intuitive dashboard to access their coursework.

The platform addresses the critical gap between content delivery and performance tracking, creating a seamless experience for both instructors and learners in online education environments.

Core Features

LearnFlow delivers three integrated modules that work together to create a complete assessment ecosystem

Classroom Management

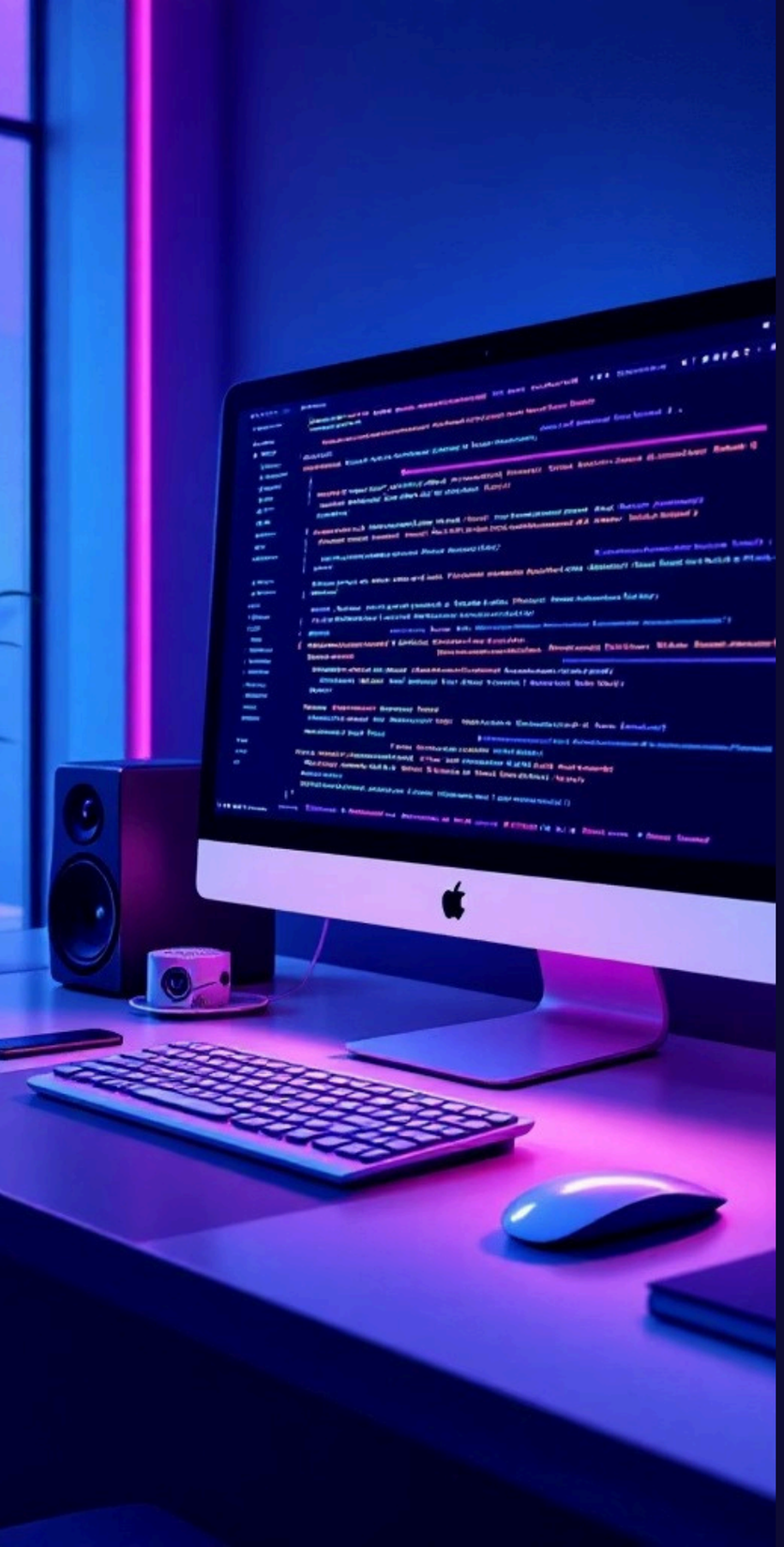
Teachers and students can effortlessly join or create classes with unique access codes. Streamlined enrollment ensures students are connected to the right courses instantly.

Dynamic Quiz Creation

Educators build assessments with flexible question types, time limits, and scoring parameters. Customize quizzes to match learning objectives and difficulty levels with ease.

Student Dashboard

Students access all enrolled classes and available quizzes from a unified interface. Track progress, view grades, and manage submissions from one centralized location.



Technical Architecture

LearnFlowis built on a robust,secure technology stack that prioritizes performance and data integrity

PHP Backend

Server-side logic handles all business operations, user authentication, and request processing. Clean, maintainable code architecture ensures scalability.

MySQL Database

Relational database management stores all user data, quiz content, and assessment results with ACID compliance for data consistency.

PDO Layer

Prepared statements and parameterized queries prevent SQL injection attacks. Secure data handling protects sensitive student information.

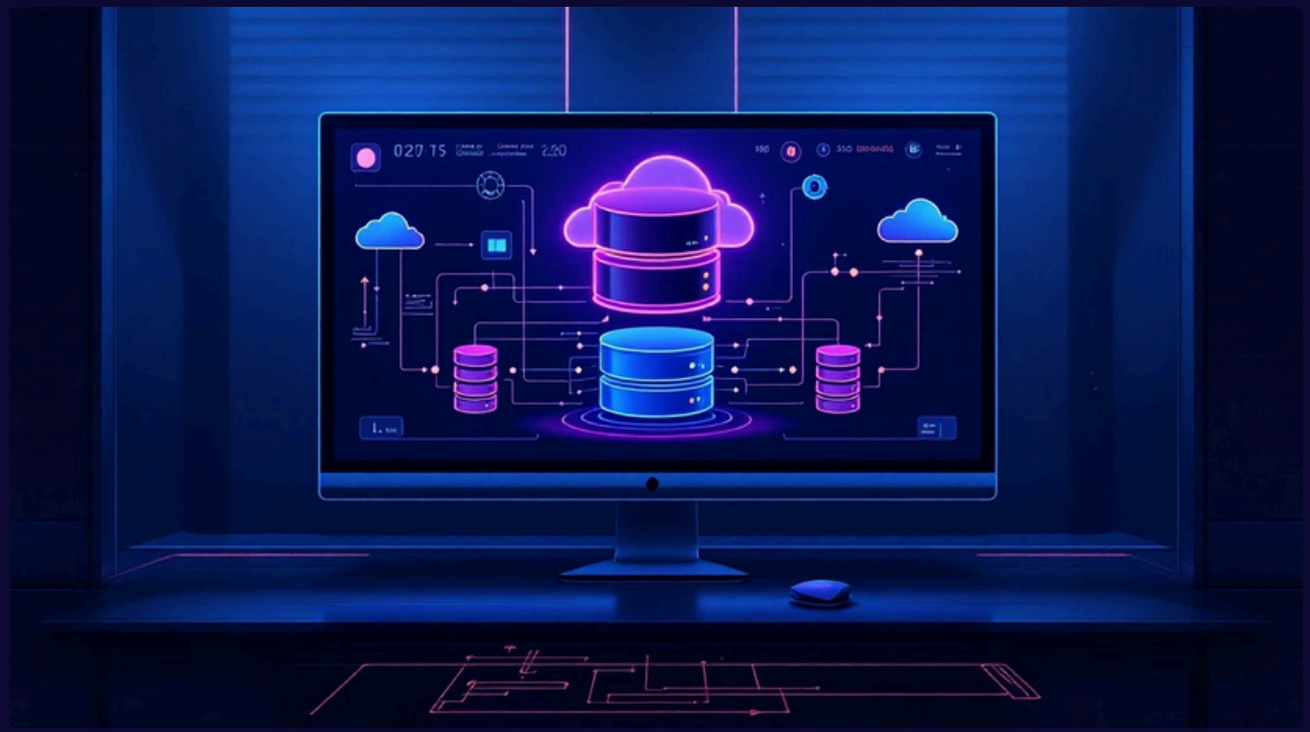
Database Design

Relational Structure

LearnFlow's database implements normalized tables that efficiently track users, quizzes, and enrollment relationships. The schema ensures data integrity while supporting complex queries for reporting and analytics.

Key relationships include:

- Users connect to multiple classes through enrollment records
- Quizzes belong to specific classes with defined parameters
- Attempt tracking maintains historical assessment data



One Attempt Only Logic: The system implements strict enforcement preventing students from submitting multiple quiz attempts. The database checks enrollment status before allowing quiz access, ensuring academic integrity.

The Gradebook System

Teachers gain comprehensive visibility into student performance through an intuitive gradebook interface that automates scoring and highlights trends.

Automated Scoring

Multiple-choice and true/false questions are graded instantly upon submission, providing immediate feedback to students and reducing teacher workload.

Performance Metrics

Track completion rates, average scores, and time-on-task across all assessments. Identify struggling students and adjust instruction accordingly.

Progress Tracking

Longitudinal data shows improvement over time with visual charts displaying score trends. Export grades for integration with institutional systems.

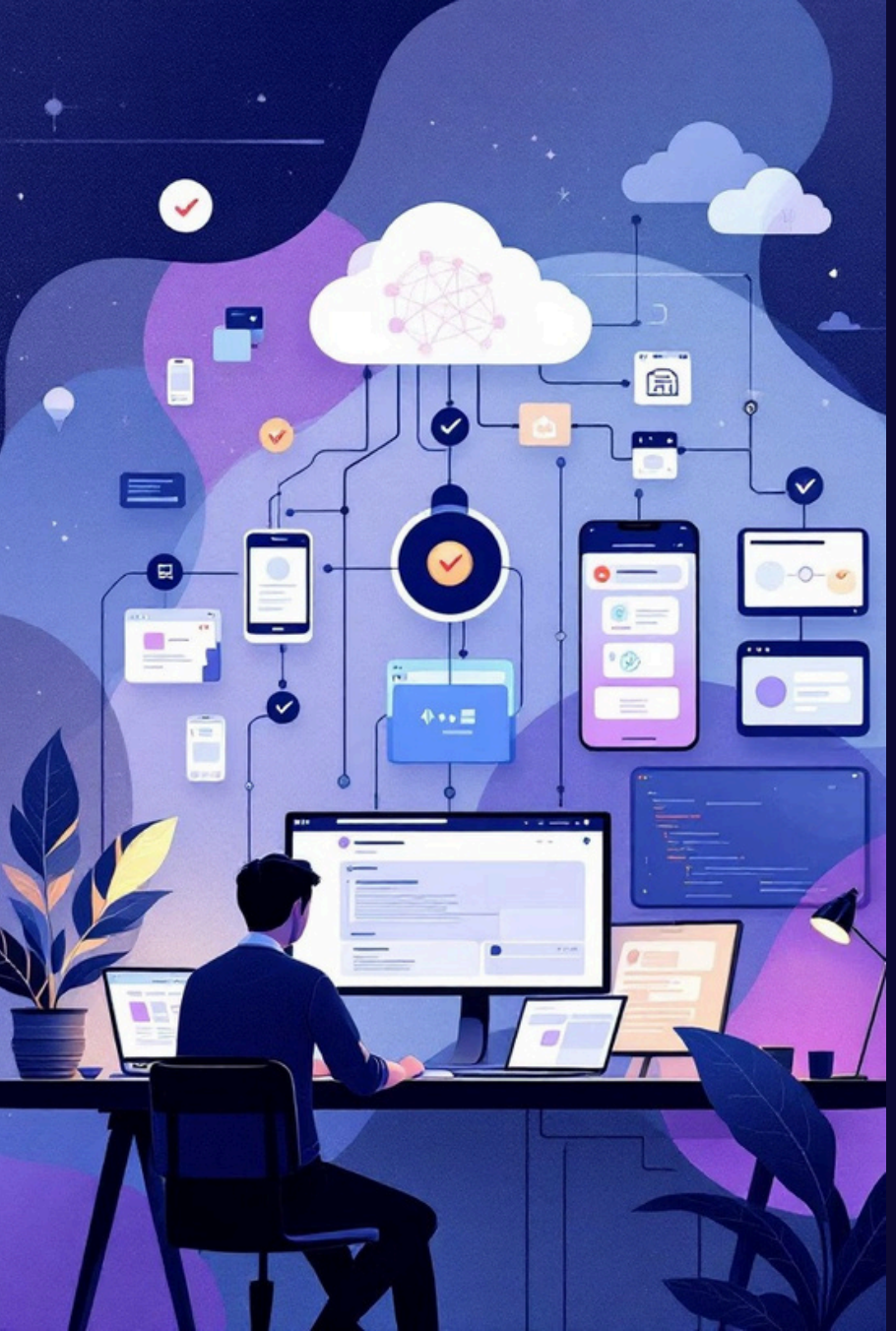
Security Measures

Protecting Academic Integrity

LearnFlow implements multiple security layersto prevent unauthorized access and ensure assessment validity. The system uses PHP sessions to maintain authenticated user state throughout the application.

Key Security Features:

- Session-based authentication prevents credential sharing
- One-attempt enforcement blocks duplicate quiz submissions
- Encrypted password storage using industry-standard hashing
- Input validation sanitizes all user-provided data
- Role-based access control separates teacher and student privileges



Implementation Highlights

01

Session Management

PHP sessions maintain user authentication state across multiple page requests, ensuring users remain logged in securely throughout their session.

02

QuizAccess Control

Before allowing quizaccess, the system verifies enrollment status and checks if an attempt has already been recorded in the database.

03

Result Recording

Uponquiz completion,scores are calculated, stored with timestamp, and associated with the specific student and quiz attempt for auditing.

Future Roadmap

LearnFlow is positioned for continuous innovation,with exciting features planned to enhance the teaching and learning experience

AI Question Generation

Intelligent algorithms automatically create quiz questions from course content, saving teachers time while maintaining assessment quality.

Real-Time Leaderboards

Engage students with competitive elements showing rankings based on scores and completion speed across classes.

File Sharing

Expand beyond quizzes to support document distribution, assignment submissions, and multimedia content sharing within classes.